



SPI FRAM GPIO Overview

FRAM Connections, Initialization and Implementation Steps

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- GPIOs Used for Connections with the FRAM
- SPI FRAM 64KBIT MB85RS640
- STM32F446RE Nucleo Board to FRAM Connections
- Practical Implementation Steps

GPIOs Used for Connections with the FRAM

Serial Clock (SCK), Master In Slave Out (MISO), Master Out Slave In (MOSI), Chip Select (CS)

- SCK Port: [GPIOB](#)
- SCK Pin: [GPIO_PIN_3](#)

- MISO Port: [GPIOB](#)
- MISO Pin: [GPIO_PIN_4](#)

- MOSI Port: [GPIOB](#)
- MOSI Pin: [GPIO_PIN_5](#)

- CS Port: [GPIOA](#)
- CS Pin: [GPIO_PIN_10](#)

- Pinout & Connectivity Options: [Nucleo Pinout](#)

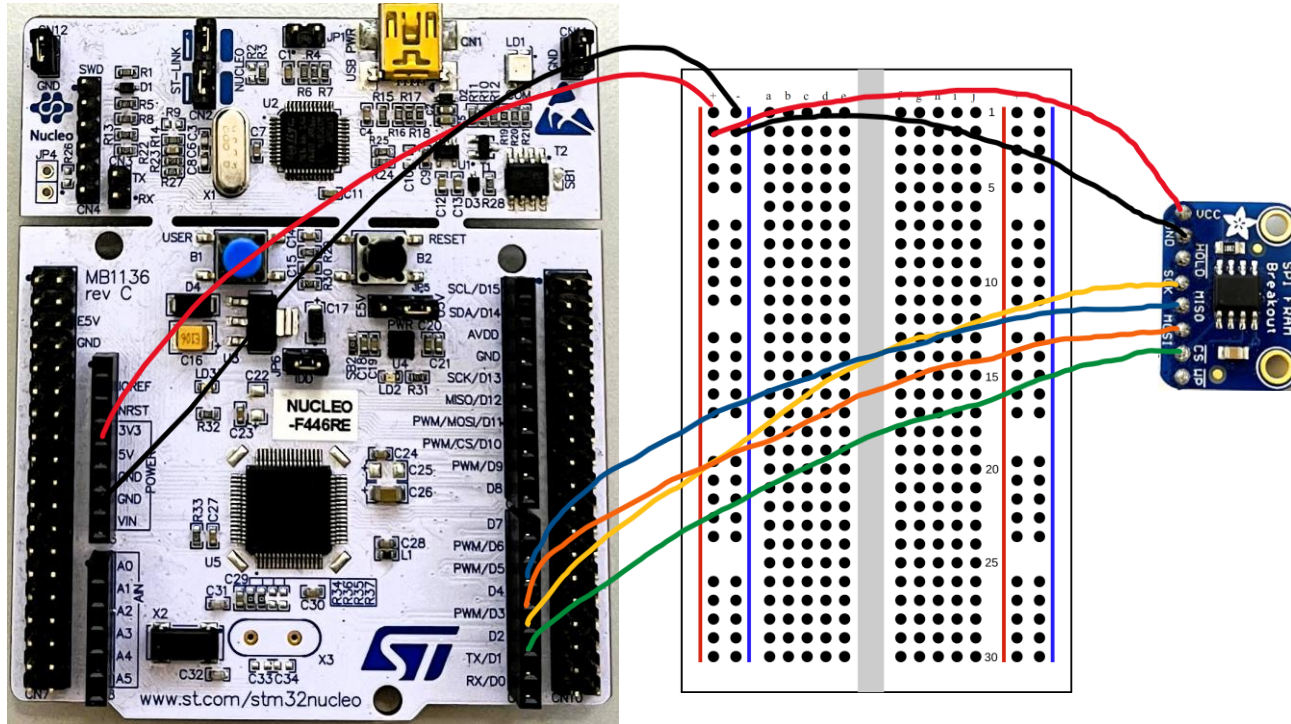
SPI FRAM 64KBIT MB85RS64

Reminder: Where to Purchase & SPI Connections

- Vendors: [Adafruit](#), [DigiKey](#)

STM32F446RE Nucleo Board to Sensors Connections

Red: **3.3V**, Black: **GND**, Yellow: **SCK**, Blue: **MISO**, Orange: **MOSI**, Green: **CS**



Practical Implementation Steps

Port/Pin Definitions

- **SPI GPIO Definitions:** Include SPI GPIO definitions in `gpio_defs.h` for SCK, MISO, MOSI and CS Port/Pins.

GPIO Initialization

- **SPI1 Alternate Function:** Configure PB3, PB4, PB5 to Alternate Function 5 (AF5) for SPI1 SCK, MISO and MOSI (refer to datasheet Table 11).
- **Chip Select (CS) Initialization:** Configure Chip Select as an output for selecting the FRAM.

Importance

- Completes GPIO initialization for reliable SPI communication for the FRAM used by the Modbus Data Manager Task.

The background features a complex network of thin, light-colored lines and small circles, forming various triangular shapes. Some of these shapes are filled with a very light yellow or green color. The overall effect is a subtle, geometric pattern that resembles a network or a molecular structure.

Next: SPI FRAM GPIO Initialization
